

Problem Statement

Business Analyst Case Study | Dataset: Olist E-Commerce (seller marketplace)

01 · PROBLEM STATEMENT

The Business Problem

Order cancellations and unfulfilled ('unavailable') orders create direct revenue loss, wasted logistics cost (reverse pickups, refund processing, repeated delivery attempts), and erode customer trust — particularly damaging for first-time or infrequent shoppers, who are more likely to churn after one bad experience. Across the dataset's 99,441 orders, 1,234 fell into this category — an overall rate of **1.24%**.

This report investigates that 1.24% rate: whether it reflects a diffuse, low-priority background rate, or whether it is concentrated in an identifiable, fixable part of the business.

Why a Low Aggregate Rate Still Deserves Investigation

1.24% looks small enough to dismiss at a glance. Before committing analysis time to this metric, it is worth being explicit about why it's still a reasonable candidate for a root-cause investigation.

1. Averages hide concentration

A flat 1.24% spread evenly across every seller and category would genuinely not be worth investigating further — there would be no lever to pull. But if that same 1.24% is actually close to 0.4% for most sellers and 2.5–3% for a specific, identifiable segment, the situation changes entirely: the problem isn't diffuse, it's concentrated, and concentrated problems are fixable problems.

Worked example: suppose 90% of orders run at a 0.4% cancellation rate, and the remaining 10% (a specific, identifiable cohort) run at 3%. The blended average across both groups works out to roughly 0.66% — low enough that a single aggregate number would not raise any alarm. Yet that 10% cohort has a rate 7.5 times the baseline, and represents a fully fixable, high-leverage target once isolated. This is precisely the kind of pattern an aggregate metric is structurally unable to reveal on its own.

2. Scale changes what "small" means

At a large marketplace's real order volumes (potentially millions per month), even a 1% rate translates into a large absolute number of failed transactions — each carrying its own cost in reverse logistics, refund processing, wasted delivery capacity, and customer support load. A percentage that looks negligible in isolation can represent a meaningful absolute cost at scale.

3. Leverage matters more than size

The objective of this analysis is not to treat 1.24% as a crisis, but to find a specific, low-cost lever that meaningfully reduces a segment's rate without requiring a company-wide policy change. A small-but-concentrated, fixable pattern is a more useful finding for a business than a large, uniform, structurally unfixable one.

4. Growth compounds the risk

New sellers are typically the fastest-growing part of a marketplace actively onboarding new supply. If that segment structurally underperforms and is also the fastest-growing part of the seller base, the aggregate cancellation rate will not stay flat over time — it will drift upward as the mix shifts toward the underperforming segment, unless the issue is identified and addressed early.

Scope of This Analysis

Item	Detail
Dataset	Olist Brazilian E-Commerce (public proxy for a seller marketplace)
Orders analyzed	99,441 orders / 112,650 order-items
Problem definition	order_status IN ('canceled', 'unavailable')
Segmentation dimensions tested	Product category, seller tenure, delivery delay
Out of scope	Refund cost modeling, customer-level churn impact, payment-method effects